Wall time per step: Splat simulation, 500k (482,409 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	48	29	21	19	OOM	11	21	203	18	228	216	35	65	156	37
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	6.3	13	13	4.0	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.0	0.8	0.8	1.9	13	13
hvg	1.0	1.1	0.8	0.7	–	3.1	4.5	0.5	0.5	19	19	19	13	7.2	7.2
pca	2.3	2.3	1.3	1.1	–	10	11	15	15	71	OOM	OOM	274	23	28
harmony	1.7	1.6	1.1	0.7	–	2.1	2.0	100	83	84	–	–	88	72	81
knn_snn	14	14	7.5	6.7	–	33	29	40	40	42	–	–	45	145	159
cluster	4.1	4.0	3.2	3.1	–	20	20	115	123	103	–	–	133	16	16
cluster_fast	14	14	3.1	2.5	n/a	n/a	n/a	196	194	197	n/a	n/a	n/a	n/a	n/a
umap	35	34	3.9	3.7	–	226	226	182	175	178	–	–	257	421	423
dge	2.3	2.3	1.8	1.9	–	454	–	22	23	1,717	–	–	260	13	13
tsne_default	252	251	223	225	–	1,315	1,319	n/a	n/a	n/a	–	–	1,390	n/a	n/a
tsne_fft	63	61	14	20	–	153	156	n/a	n/a	n/a	–	–	156	n/a	n/a
tsne_quick	42	48	27	23	n/a	n/a	n/a	132	133	135	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	109	88	41	37	failed	766	317	681	481	2,448	failed	failed	1,140	866	777

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.